

The process we use to write software with Go is straightforward:

- Gather requirements
- Find a solution
- Write source code to implement the solution
- Compile the source code into an executable
- Run and test the program to make sure it works

This process is iterative and the steps usually overlap.



Figure 1: Go's mascot: Gopher

What is Abot?

Abot is a digital assistant framework written in the Go programming language. Abot's framework is designed to help users in building their own assistant and customising it according to their needs—whether it's a computer that answers phones, intelligently routes calls and schedules your business travel, or just a better take on Siri that orders Ubers and does your laundry.

Abot exposes a simple HTTP API, so you can easily connect it to send, process and respond to texts, emails and more.

How the framework of Abot works

The brain of the Abot tool consists of three parts:

- An API (application program interface) that accepts natural language (human language) input.
- A state machine for tracking grammar and context across inputs to enable chaining of commands.
- A router to select the correct plugin that sends inputs based on the present commands and past context.

Abot combines these three parts and libraries to hasten the process of coding one's personal digital assistant.

Abot processes, routes and responds to every message that it receives. Actually deciding what to say is the role of plugins. Let's take a look at an example:

1. A user sends the following message via the console, SMS, email, etc: 'Show me Chinese restaurants nearby.'
2. Abot pre-processes the message, breaking it into key components:
 - a. Commands: [Show]
 - b. Objects: [me, Chinese restaurants nearby]
3. Abot routes this message to the respective plugin:
 - a. Route: [find_chinese]
 - b. Plugin: [restaurant]
4. The plugin then generates a response: 'Sure, how does Sun Shi sound? It's nearby.'
5. Abot sends the response to the user.

Abot sends the response through the same channel the user chose, so if you send Abot a text, it will respond in text. Thus, every message goes from *User* --> *Abot* --> *Plugin* --> *Abot* --> *User*.

How to build your own personal assistant

As mentioned earlier, Abot is a digital assistant framework that is based on the Go programming language. It requires some prerequisites like Go (version 1.6 or later) and PostgreSQL (version >= 9.5). Once you have ensured that you have these dependencies, we can move forward to running the Abot server on your terminal.

1. Fetch Abot using *go get*:

```
go get github.com/itsabot/abot
cd $GOPATH/src/github.com/itsabot/abot
go get github.com/robfig/glock
```

2. Set the Glock binary path:

```
export PATH=$PATH:/home/ubuntu/workspace/bin/
```

3. Run the setup script, by passing it in your PostgreSQL credentials/host, if needed:

```
cd $GOPATH/src/github.com/itsabot/abot
cmd/setup.sh [username[:password]@host[:port]]
```

If you don't pass anything to the script, Postgres will, by default, set the host = 127.0.0.1, port = 5432 and username = postgres.

4. Once the script is completed, run the following command:

```
$ abot server
```

And then visit Abot at *localhost:4200*.

Once you have everything installed and running, you can start communicating with Abot. To do so locally, use the command *\$ abot* at the console. But before that, you'll need to sign up. To do so, go to *http://localhost:4200* and click on 'Sign up' in the header. Fill in the appropriate information, and remember the phone number you use (which should be in the format of +918889992221 — no spaces, and with a leading +91).

Once you've created an account, type the following code:

```
$ abot console +918889992221
> Hi
Hi there. :)
```

Replace the number with the one you used to sign up. Abot won't do much at this point; we will need to add plugins to add more functionalities.